Relativistic Quantum Mechanics as a Forced Structural Consequence of Event-Density Primitives

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Abstract

We show that relativistic quantum mechanics — including the Klein–Gordon equation, the spin–statistics correspondence, the Clifford-algebra spinor frame, and the Dirac equation — emerges as a *forced structural consequence* of the Event-Density (ED) primitive stack, rather than as a set of independent postulates. Building on the Phase-1 derivation of non-relativistic quantum mechanics from the participation measure $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$, we extend the program to the relativistic regime via three connected structural theorems. **Theorem R1** (spin–statistics) establishes $\eta = (-1)^{2s}$ at primitive level via the chain: exchange dichotomy $\to \pi_1(Q_2) = \mathbb{Z}_2 \to$ exchange-rotation generator identity $\to$ Clifford frame $\to$ minimal-bilinear-pairing closure. **Theorem R2** (Cl(3,1) uniqueness) shows that the real Clifford algebra with anticommutator $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ is the unique algebraic structure compatible with Lorentz covariance, half-integer representation content, and the rotational double cover; pure commutation and pure anticommutation alternatives are ruled out. **Theorem R3** (Dirac emergence) derives the Dirac equation $(i\gamma^\mu\partial_\mu - mc/\hbar)\Psi = 0$ as the unique first-order Lorentz-covariant linear equation on the Cl(3,1) spinor module, with positive-definite conserved current $j^\mu = \bar{\Psi}\gamma^\mu\Psi$, gauge-covariant minimal coupling, and tree-level $g = 2$ from non-relativistic reduction. ED inherits dimensional constants $(\hbar, c, m, q)$; it derives the *form* of relativistic quantum equations.

1 Introduction

The mathematical structure of quantum mechanics — wavefunctions on Hilbert space, unitary evolution, the Born rule, exchange statistics, spinor representations of the Lorentz group, and the first-order Dirac equation — is conventionally introduced as a layered set of independent postulates. Each layer (kinematic, dynamical, statistical, relativistic, gauge-theoretic) is justified primarily by its empirical success rather than by derivation from a more fundamental ontology.

The Event-Density (ED) program offers an alternative: to build quantum mechanics as a *forced structural consequence* of a small primitive stack governing micro-events, chains, bandwidth content, and commitment dynamics on a 3+1-dimensional event manifold. Phase-1 of this program [1] established that non-relativistic single-particle quantum mechanics — the Schrödinger equation, the Born rule, Bell–CHSH violations saturating the Tsirelson bound, and the Heisenberg uncertainty relation — emerges from the participation measure

$$P_K(x,t) = \sqrt{b_K(x,t)} \cdot e^{i\pi_K(x,t)}, \tag{1}$$

combined with the four-band bandwidth decomposition (Primitive 04) and commitment dynamics (Primitives 10–11). The squared-amplitude exponent $b_K = |P_K|^2$ is *definitional*, not postulated; the Born rule's $|\psi|^2$ is therefore not an independent axiom. The same exponent-2 structure threads through Madelung's $\sqrt{\rho}$, Schrödinger's kinetic term, sublinear bandwidth composition, and Heisenberg's $(\Delta x \cdot \Delta p)^2$, yielding ED's strongest positive structural unification.

Phase-2 extends the program. Arc R, the subject of the present paper, derives **relativistic single-chain quantum mechanics** at primitive level. Three structural theorems are established.

Arc M [2] (chain-mass structure) and Arc Q [3] (QFT extension) build on Arc R; we summarise their relationship in Section 6.

The motivation for deriving rather than postulating is threefold. *First*, structural derivation reduces the axiomatic surface area of physics: postulates that turn out to be theorems carry no independent empirical content and can be eliminated. *Second*, structural derivation exposes the precise inputs each piece of quantum theory requires; this clarifies which features are inherited (numerical constants, specific particle content) versus structural (form of equations, classification of representations). *Third*, structural derivation enables systematic extension: if non-relativistic quantum mechanics is a thin-participation limit of a richer primitive structure, the same primitives extend to relativistic and field-level regimes by relaxing the limit, rather than by introducing new postulates.

The Arc R results consist of three theorems plus extensive supporting structural content:

- **R1 (Spin–statistics):** $\eta = (-1)^{2s}$ derived from primitive-level exchange topology and frame structure;
- **R2 (Cl(3,1) uniqueness):** the real Clifford algebra with anticommutator $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ is the unique algebraic frame compatible with the Lorentz / spin / topology constraints;
- **R3 (Dirac emergence):** the Dirac equation is the unique first-order Lorentz-covariant equation on the Cl(3,1) spinor module, with conserved current and tree-level $g = 2$ as automatic consequences.

ED's primitive-level account does *not* predict the numerical values of dimensional constants ($\hbar$, c) or rule-type-specific parameters (m, q); these are inherited via the Dimensional Atlas. The clean separation between *form* (forced structurally) and *value* (inherited empirically) is a load-bearing methodological principle that recurs throughout Arc R and continues into Arc M and Arc Q.

2 Background

2.1 The ED primitive stack

Event Density is built on thirteen primitives. The primitives relevant to Arc R are summarised below:

- **Primitive 01 (micro-event):** discrete events on the 3+1-dimensional ambient manifold; substrate for all participation.
- **Primitive 02 (chain):** worldlines γ_K of persistent participation.
- **Primitive 03 (participation):** act of a chain entering a micro-event with non-zero amplitude.
- **Primitive 04 (bandwidth):** four-band decomposition $b_K = b_K^{\text{int}} + b_K^{\text{adj}} + b_K^{\text{env}} + b_K^{\text{com}}$.
- **Primitive 06 (four-gradient):** ∂_μ on the event manifold; carries Lorentz covariance.
- **Primitive 07 (rule-type):** discrete classification by structural levers L1–L4.
- **Primitive 09 (polarity phase):** π_K of the participation measure.
- **Primitive 10 (individuation):** threshold separating distinct chains.
- **Primitive 11 (commitment):** dynamical update producing micro-event events along γ_K.
- **Primitive 13 (relational timing):** proper time τ_K along γ_K.

2.2 The participation measure

The Phase-1 participation measure

$$P_K(x^\mu) = \sqrt{b_K(x^\mu)} \cdot e^{i\pi_K(x^\mu)} \in \mathbb{C} \tag{2}$$

fuses Primitive 04 (bandwidth amplitude) with Primitive 09 (polarity phase). Stage R.1 promotes this object to a Lorentz-covariant scalar field on the ambient 3+1D event manifold, with $x^\mu = (ct, \mathbf{x})$. Lorentz covariance is forced by Primitive 06.

2.3 Rule-type ontology

A rule-type τ is specified by four levers from Primitive 07:

- **L1:** bandwidth partition pattern w_τ^X across the four bands;
- **L2:** internal-index content (Lorentz representation (j_L, j_R) plus gauge index);
- **L3:** interface content (Fierz class Γ_τ or analogue);
- **L4:** statistics class (Case P bosonic / Case R fermionic).

Stage R.2 of Arc R produces the spin–statistics correspondence connecting L4 to L2.

2.4 Commitment and individuation

Primitive 11 supplies the dynamical update along chain worldlines: discrete events in which a chain "commits" — depositing a definite outcome on the event manifold. Primitive 10 sets the threshold at which two same-type chains remain distinguishable; for Case-R rule-types, individuation forbids same-type coincidence, leading to antisymmetry and Pauli exclusion. The interplay of these two primitives is central to the spin–statistics theorem of Section 4.

3 Arc R Stage R.1 — Scalar Relativistic Quantum Mechanics

3.1 Klein–Gordon equation

The Stage R.1 derivation begins with the Lorentz-covariant participation measure on the $(0, 0)$ representation. For a free scalar rule-type, requiring (i) Lorentz-scalar form of the dynamical equation, (ii) translation invariance, and (iii) reduction to the non-relativistic Schrödinger limit yields the Klein–Gordon equation

$$\left(\Box + \frac{m^2c^2}{\hbar^2}\right)\Psi = 0, \tag{3}$$

where $\Box = \eta^{\mu\nu}\partial_\mu\partial_\nu$. The mass term $m^2c^2/\hbar^2$ enters as the Casimir eigenvalue of the Poincaré algebra acting on the rule-type's participation module; numerical m is rule-type data inherited from the Dimensional Atlas.

3.2 Minimal coupling

Local-phase invariance $\Psi \to e^{iq\alpha(x)/\hbar}\Psi$ forces the connection $A_\mu \to A_\mu - \partial_\mu\alpha$. The covariant derivative

$$D_\mu = \partial_\mu + \frac{iq}{\hbar}A_\mu \tag{4}$$

ensures $D_\mu \Psi$ transforms covariantly. The interacting Klein–Gordon equation

$$\left(D_\mu D^\mu + \frac{m^2 c^2}{\hbar^2} \right) \Psi = 0 \tag{5}$$

is gauge-covariant. The U(1) gauge structure is therefore *forced* at primitive level by local-phase invariance, not postulated.

3.3 Conserved current

Standard manipulation yields the conserved Klein–Gordon current

$$j^\mu = \frac{i\hbar}{2m} \left[\Psi^* D^\mu \Psi - \Psi (D^\mu \Psi)^* \right], \tag{6}$$

satisfying $\partial_\mu j^\mu = 0$ on solutions. The j^0 component is sign-indefinite, signalling the well-known negative-probability pathology of Klein–Gordon for matter fields. This pathology is resolved by Stage R.3's Dirac construction (Section 5).

3.4 Stage R.1 status

Stage R.1 produces:

- Lorentz-covariant scalar participation measure (FORCED);
- Klein–Gordon equation (FORCED);
- Minimal coupling and U(1) gauge structure (FORCED);
- Scalar conserved current (FORCED, sign-indefinite for matter — superseded by spinor case).

Numerical m, q, $\hbar$, c are inherited.

4 Arc R Stage R.2 — Rule-Type Taxonomy and Spin–Statistics

Stage R.2 establishes the structural framework for spin and statistics. We summarise the five-memo sub-program (R.2.1 exchange, R.2.2 Lorentz representations, R.2.3 double-cover topology, R.2.4 Clifford algebra, R.2.5 synthesis) and state the spin–statistics theorem.

4.1 Exchange dichotomy (R.2.1)

For two same-type chains K_A, K_B, the exchange operation E_{AB} swaps participation labels. Involutive structure ($E_{AB}^2 = \mathrm{id}$) and mutual substitutability of same-type chains restrict the exchange factor $\eta(\tau)$ to

$$\eta(\tau) \in \{+1, -1\}. \tag{7}$$

Two rule-type categories emerge: **Case P** (bandwidth-sharing-permissive, $\eta = +1$, symmetric joint participation) and **Case R** (bandwidth-sharing-restrictive, $\eta = -1$, antisymmetric, vanishing-on-coincidence, Pauli exclusion).

4.2 Lorentz representations (R.2.2)

Finite-dimensional representations of the complexified Lorentz algebra $\mathfrak{so}(3,1)_{\mathbb{C}} \cong \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}$ are classified by pairs (j_L, j_R) with $j_L, j_R \in \{0, 1/2, 1, 3/2, \dots\}$ and dimension $(2j_L+1)(2j_R+1)$. The spin quantum number s runs through $|j_L - j_R| \leq s \leq j_L + j_R$ via the Pauli–Lubanski Casimir. The exhaustive ladder

$$s \in \{0, 1/2, 1, 3/2, 2, \dots\} \tag{8}$$

is rigid in 3+1D by Lie-algebraic theorem.

Integer-spin representations descend to $\mathrm{SO}^+(3,1)$; half-integer representations require the double cover $\mathrm{SL}(2,\mathbb{C})$.

4.3 Configuration-space topology (R.2.3)

The fundamental group of the two-identical-chain configuration space $Q_2 = (\mathbb{R}^3 \times \mathbb{R}^3 \setminus \Delta)/S_2$ in 3+1D evaluates to

$$\pi_1(Q_2) = \mathbb{Z}_2. \tag{9}$$

Centre-of-mass + relative-coordinate decomposition: the relative-coordinate space $(\mathbb{R}^3 \setminus \{0\})/\mathbb{Z}_2 \simeq \mathbb{R}P^2$, with $\pi_1(\mathbb{R}P^2) = \mathbb{Z}_2$.

Consequences:

1. Anyonic exchange phases are *forbidden* in 3+1D — only $\eta \in \{\pm 1\}$ admissible.
2. The exchange-path generator equals the 2π-rotation generator under the SO(3)-equivariant embedding of the relative coordinate.

4.4 Clifford algebra uniqueness (R.2.4)

Half-integer representations require an algebraic frame in which 0 and 2π rotations are distinguishable. Consider the structural problem: find the unique real finite-dimensional associative algebra $\mathcal{A}$ generated by elements e_μ ($\mu = 0, 1, 2, 3$) satisfying:

(P1) Lorentz tangent-space compatibility: $\{e_\mu\}$ transforms as a four-vector;

(P2) Metric compatibility: symmetric pairings reproduce $\eta_{\mu\nu}$;

(P3) Half-integer representation realisation;

(P4) Minimal dimension.

Pure commutation ($e_\mu e_\nu = e_\nu e_\mu$) fails (P3); pure anticommutation ($e_\mu e_\nu = -e_\nu e_\mu$ strictly) fails (P2). The unique solution is the **anticommutator** structure

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \cdot \mathbb{1},} \tag{10}$$

defining the real Clifford algebra $\mathrm{Cl}(3,1) \cong M_4(\mathbb{R})$ of dimension 16. The Lorentz generators on the spinor module are

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu], \tag{11}$$

generating the $\mathrm{SL}(2,\mathbb{C})$ action. Direct computation yields $D(R(2\pi)) = -\mathbb{1}$ on the spinor module — the rotational double-cover sign emerges automatically from the half-angle in $\exp(-i\theta\sigma^{12}/2)$.

4.5 Spin–statistics synthesis (R.2.5)

Theorem 4.1 (Spin–Statistics, R1)**.** *For any ED rule-type in 3+1D with Lorentz-covariant internal index structure, the exchange sign η and the spin quantum number s are related by*

$$\boxed{\eta = (-1)^{2s},} \tag{12}$$

with η FORCED to lie in $\{\pm 1\}$ by $\pi_1(Q_2) = \mathbb{Z}_2$ and s FORCED to lie in $\{0, 1/2, 1, 3/2, 2, \dots\}$ by Lorentz representation theory. Integer-spin rule-types are Case P (bosonic); half-integer-spin rule-types are Case R (fermionic).

Sketch. The chain runs through six steps:

1. Exchange dichotomy $\eta \in \{+1, -1\}$ from rule-type structure (R.2.1).
2. Topological upgrade $\pi_1(Q_2) = \mathbb{Z}_2$ from spatial dimension 3 + identical-chain exchange (R.2.3).
3. Geometric theorem: exchange-path generator = 2π-rotation generator (R.2.3).
4. Lorentz-representation ladder: half-integer requires $\mathrm{SL}(2, \mathbb{C})$ (R.2.2).
5. Algebraic frame: $\mathrm{Cl}(3, 1)$ with $D(R(2\pi)) = -\mathbb{1}$ on spinor module (R.2.4).
6. Bilinear-pairing closure: Primitive 10 individuation supplies the bilinear $\bar{\Psi}\Gamma\Psi$ coupling between chain indices, fixing the rule-type-frame coupling (R.2.5).

The minimal-bilinear closure step uses Primitive 10's two-chain individuation pairing made dynamical by Primitive 11. No CANDIDATE assumptions remain; the theorem is FORCED unconditionally at primitive level. □

Dependencies on primitives. 02, 03, 06, 07, 10, 11, 13.

4.6 Cl(3,1) uniqueness as a separate theorem

We isolate the algebraic-frame content of Section 4.4 as a theorem:

Theorem 4.2 (Cl(3,1) Uniqueness, R2)**.** *The unique real finite-dimensional associative algebra compatible with (P1) Lorentz tangent-space compatibility, (P2) metric compatibility, (P3) half-integer representation realisation, and (P4) minimal dimension is the real Clifford algebra* $\mathrm{Cl}(3, 1)$ *with $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \cdot \mathbb{1}$. Pure commutation fails (P3); pure anticommutation (strict) fails (P2). The anticommutator is structurally mandatory.*

Sketch. Suppose $e_\mu e_\nu = e_\nu e_\mu$ for all μ, ν. Then (P2) reduces to $e_\mu e_\nu = (\eta_{\mu\nu}/2)\mathbb{1}$, forcing $e_\mu = 0$ for $\mu \neq \nu$ and rigid scalars on the diagonal. The resulting algebra does not support a half-integer $\mathrm{SL}(2, \mathbb{C})$ representation; (P3) fails.

Suppose $e_\mu e_\nu = -e_\nu e_\mu$ strictly. Then $e_\mu^2 = 0$ for all μ, contradicting (P2) on the diagonal.

The remaining possibility is $e_\mu e_\nu + e_\nu e_\mu = 2\eta_{\mu\nu}\mathbb{1}$, the Clifford-algebra defining relation, which generates $\mathrm{Cl}(3, 1)$ uniquely up to isomorphism, with dimension $2^4 = 16$ and basis $\{\mathbb{1}, \gamma^\mu, \gamma^{\mu\nu}, \gamma^{\mu\nu\rho}, \gamma^5\}$. □

Dependencies on primitives. Primitive 06 (Lorentz covariance + metric) plus the half-integer-representation requirement from Theorem R1.

5 Arc R Stage R.3 — Dirac Emergence

5.1 Square-root factorisation

The Stage R.1 Klein–Gordon equation applied component-wise to a spinor field Ψ_α does not use the γ^μ generators non-trivially. To use the algebraic frame, a first-order Lorentz-covariant equation built from γ^μ, ∂_μ, and a mass scale is required.

Compute:

$$\begin{aligned}(i\gamma^\mu\partial_\mu)(i\gamma^\nu\partial_\nu) &= -\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu \\ &= -\tfrac{1}{2}\{\gamma^\mu,\gamma^\nu\}\partial_\mu\partial_\nu \\ &= -\eta^{\mu\nu}\partial_\mu\partial_\nu\cdot\mathbb{1} = -\Box.\end{aligned} \tag{13}$$

Therefore

$$(i\gamma^\mu\partial_\mu - mc/\hbar)(i\gamma^\mu\partial_\mu + mc/\hbar) = -\Box - m^2c^2/\hbar^2. \tag{14}$$

A spinor satisfying the first-order equation

$$\boxed{(i\gamma^\mu\partial_\mu - mc/\hbar)\,\Psi = 0} \tag{15}$$

automatically satisfies Klein–Gordon component-wise. This is the **Dirac equation**.

5.2 Uniqueness of the first-order form

Three independent arguments force the first-order form for Case-R rule-types:

1. **Use of the Cl(3,1) structure.** Theorem R2 establishes γ^μ as primitive-level frame generators. A second-order equation leaves this structure inert. Equation (15) is the unique Lorentz-covariant first-order linear operator on Ψ constructible from γ^μ, ∂_μ, and a mass scale.

2. **Single time-derivative / positivity.** Klein–Gordon as a second-order spinor equation gives an indefinite inner product. The first-order Dirac form yields a positive-definite current (Section 5.3).

3. **Half-integer representation content.** For the $(1/2,0)\oplus(0,1/2)$ Dirac module, the minimal Lorentz-covariant first-order equation is unique up to equivalence: it is the Dirac equation.

Theorem 5.1 (Dirac Emergence, R3)**.** *Given Stage R.1 (Klein–Gordon Casimir) and Stage R.2 (Cl(3,1) frame, half-integer representation content), the unique first-order Lorentz-covariant linear dynamical equation for Case-R rule-type participation measures is*

$$(i\gamma^\mu\partial_\mu - mc/\hbar)\,\Psi = 0. \tag{16}$$

No additional postulates are required. The equation is structurally FORCED.

Sketch. Lorentz-covariance plus first-order linear structure built from $\gamma^\mu, \partial_\mu, m$ admits exactly the family $a\cdot i\gamma^\mu\partial_\mu\Psi + b\cdot(mc/\hbar)\Psi = 0$ with a, b scalars. Squaring must reproduce Klein–Gordon component-wise (mass-shell consistency); this forces $a = 1, b = -1$ up to overall normalisation. The Cl(3,1) anticommutator $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$ ensures that squaring eliminates the cross term $\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu$ via the symmetry of $\partial_\mu\partial_\nu$. □

Dependencies on primitives. All Arc R stages: R.1 Casimir + R.2 Cl(3,1) frame + R.2 spin–statistics for the Case-R identification.

5.3 Conserved current

Multiplying the Dirac equation by $\bar{\Psi} = \Psi^\dagger \gamma^0$ and combining with the adjoint equation, the mass terms cancel and we obtain $\partial_\mu(\bar{\Psi}\gamma^\mu\Psi) = 0$, so

$$j^\mu \equiv \bar{\Psi}\gamma^\mu\Psi \tag{17}$$

is conserved. The time component

$$j^0 = \bar{\Psi}\gamma^0\Psi = \Psi^\dagger\gamma^0\gamma^0\Psi = \Psi^\dagger\Psi = \sum_\alpha |\Psi_\alpha|^2 \geq 0 \tag{18}$$

is **positive-definite**, in contrast to the indefinite Klein–Gordon current. The Dirac construction therefore resolves the negative-probability pathology of Klein–Gordon for spin-1/2 matter fields. Positivity is structurally tied to the first-order form via the spinor-module structure.

5.4 Minimal coupling and gauge invariance

The Stage R.1 minimal-coupling procedure applies unchanged on the spinor module:

$$(i\gamma^\mu D_\mu - mc/\hbar)\Psi = 0, \qquad D_\mu = \partial_\mu + \frac{iq}{\hbar}A_\mu. \tag{19}$$

Under local U(1), $\Psi \to e^{iq\alpha/\hbar}\Psi$, $A_\mu \to A_\mu - \partial_\mu\alpha$, and $D_\mu\Psi \to e^{iq\alpha/\hbar}D_\mu\Psi$. The interacting Dirac equation is gauge-covariant.

5.5 Non-relativistic reduction and $g = 2$

In the non-relativistic limit, write $\Psi = (\varphi, \chi)^T \cdot e^{-imc^2t/\hbar}$ and solve the small-component equation algebraically. Substituting back and keeping leading-order terms yields the **Pauli equation**:

$$i\hbar\partial_t\varphi = \left[\frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + qA^0 - \frac{q\hbar}{2m}\boldsymbol{\sigma}\cdot\mathbf{B}\right]\varphi. \tag{20}$$

The Zeeman coefficient identifies the gyromagnetic ratio

$$\boxed{g_{\text{Dirac}} = 2,} \tag{21}$$

emerging without any parameter tuning from the Cl(3,1) frame and the first-order form. Higher-order QED corrections $g - 2 \approx \alpha/\pi + \dots$ are loop-level content (Arc Q [3] handles these); the *tree-level* $g = 2$ is a structural prediction of Arc R.

Dropping spin-dependent terms (projecting onto a spinless participation) yields the Schrödinger equation, recovering the Phase-1 non-relativistic limit and closing a consistency loop across the entire QM-emergence program.

6 Discussion

6.1 What Arc R forces vs what it leaves open

Arc R forces, at primitive level:

- Lorentz-covariant scalar participation measure;
- Klein–Gordon equation for spin-0 rule-types;
- Minimal coupling and U(1) gauge structure from local-phase invariance;

- Spin–statistics theorem $\eta = (-1)^{2s}$ (Theorem R1);
- (j_L, j_R) representation classification with exhaustive spin ladder;
- Anyon prohibition in 3+1D (from $\pi_1(Q_2) = \mathbb{Z}_2$);
- Cl(3,1) frame uniqueness with $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ (Theorem R2);
- $D(R(2\pi)) = -\mathbb{1}$ on the spinor module;
- Dirac equation as unique first-order Lorentz-covariant equation on the spinor module (Theorem R3);
- Positive-definite conserved current $j^\mu = \bar{\Psi}\gamma^\mu\Psi$;
- Tree-level Dirac $g = 2$ from non-relativistic reduction;
- Phase-1 consistency: Schrödinger emerges as the spinless non-relativistic limit.

Arc R inherits (does not predict):

- Numerical $\hbar$, c;
- Per-rule-type mass m, charge q;
- Specific spin assignment per rule-type;
- Choice between Dirac, Weyl, and Majorana reality structure for any specific Case-R rule-type;
- The number of rule-types occupying any particular structural slot.

The form/value separation is preserved throughout. ED's primitive structure determines the *form* of relativistic quantum equations and the *classification* of admissible spin/statistics structures; numerical content is inherited via the Dimensional Atlas.

6.2 Relationship to Arc M

Arc M [2] addresses the chain-mass problem: whether ED primitives constrain the numerical mass values of rule-types. Building on Arc R's spin–statistics framework and Cl(3,1) frame, Arc M constructs a candidate bandwidth-signature

$$\sigma_\tau = \hbar \cdot \sqrt{\sum_X w_\tau^X \cdot \langle (\partial_\mu \ln b_\tau^X)(\partial^\mu \ln b_\tau^X) \rangle_\tau} \tag{22}$$

as a Lorentz-scalar functional of rule-type bandwidth content. Arc M closes "H1-dominant": σ_τ's Lorentz-scalar form is FORCED, but mass values, ratios, and hierarchies are INHERITED. ED fixes the *structure* of mass but not its *values*. Arc R supplies the foundational framework for Arc M's analysis.

6.3 Relationship to Arc Q

Arc Q [3] extends ED to the QFT regime: gauge-group structure, interaction vertices, Higgs-like SSB, radiative corrections, generations, second quantisation, and vacuum structure. Three Arc Q theorems extend the structural framework:

- **GRH unconditional forced** — at least one massless Case-P $(1/2, 1/2)$ gauge rule-type exists structurally.

- **Canonical (anti-)commutation forced** — second-quantisation derived directly from Arc R's spin–statistics + Primitive 10 individuation.
- **UV-FIN forced at primitive level** — Primitive 01 event-discreteness + Primitive 13 finite intervals + Primitive 04 bounded bandwidth jointly guarantee primitive-level UV finiteness; renormalisation REFUTED as fundamental.

Arc Q's results presuppose Arc R's framework: the Cl(3,1) frame supplies the vertex catalogue's Fierz basis; the spin–statistics theorem forces the (anti-)commutation structure; the Dirac equation supplies the matter-sector dynamics for QFT loop calculations.

6.4 How Arc R sets the stage for the QFT extension

Arc R closes the relativistic single-chain quantum sector and supplies the structural prerequisites for Arc Q:

- The Cl(3,1) frame from Theorem R2 is the algebraic backbone of the Arc Q vertex catalogue.
- Theorem R1's spin–statistics directly forces Arc Q's canonical (anti-)commutation relations.
- The Dirac equation of Theorem R3 supplies the matter-sector dynamics for Arc Q's interaction-vertex analysis.
- The minimal-coupling derivation extends to non-Abelian gauge groups without additional structural input.

Together, Arc R + Arc Q yields a structurally complete, UV-finite quantum field framework whose form is forced and whose numerical content is inherited.

7 Conclusion

Arc R derives relativistic single-chain quantum mechanics as a forced structural consequence of the ED primitive stack. Three structural theorems carry the load: spin–statistics (Theorem R1), Cl(3,1) uniqueness (Theorem R2), and Dirac emergence (Theorem R3). Together they reduce the conventional independent postulates of relativistic quantum theory — the form of the wave equation, the spin–statistics correspondence, the spinor frame, the gauge-covariant minimal coupling, and the conserved current — to consequences of a small primitive stack on a 3+1-dimensional event manifold.

The methodological framing — *form* FORCED, *value* INHERITED — distinguishes ED's contribution from numerical-prediction frameworks. ED does not predict the numerical value of $\hbar$, the electron mass, or the fine-structure constant. It predicts the *structural form* of the equations these constants appear in: the Klein–Gordon and Dirac equations, the gauge-covariant derivative, the conserved current, the spin ladder, the exchange-statistics dichotomy, and the Cl(3,1) algebraic frame.

Arc R's tree-level Dirac $g = 2$ emerges from the Cl(3,1) frame and the first-order form without parameter tuning. Its positive-definite conserved current resolves the Klein–Gordon negative-probability pathology for spin-1/2 fields. Its anyon prohibition follows from $\pi_1(Q_2) = \mathbb{Z}_2$ — a topological theorem about 3+1-dimensional configuration space.

These structural results ground the further extensions of Arc M (chain-mass) and Arc Q (QFT). Together, Phase-2 of the ED program (Arc R + Arc M + Arc Q) yields a structurally complete, UV-finite quantum field framework. The clean separation between structural prediction and empirical inheritance is preserved at every layer.

Phase-3 directions — ED to general-relativistic coupling, cosmological structure via the UV-FIN reframing of cosmological-Λ, possible high-energy empirical signatures near primitive event-discreteness scales — are unblocked by the Phase-2 closure of which Arc R is the relativistic foundation.

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